VAR:

x_1: QUANTITÀ DI PRODOTTO X IN TONN.
x_2: " " Y " "

VINCOLI:

- $6x_1 + 7x_2 \leq 60$
- $6x_1 + 3x_2 \leq 40$
- $x_1 \geq 2$
- $x_1, x_2 \geq 0$

LP:

$$\max z: x_1 + x_2$$
$$6x_1 + 7x_2 \leq 60$$
$$6x_1 + 3x_2 \leq 40$$
$$x_1 \geq 2$$
$$x_1, x_2 \geq 0$$

STANDARD:

$$\min(-z): -x_1 - x_2$$
$$6x_1 + 7x_2 + x_3 = 60$$
$$6x_1 + 3x_2 + x_4 = 40$$
$$x_1 - x_5 = 2$$
$$x_1, \dots, x_5 \geq 0$$

0	-1	-1	0	0	0
60	6	7	1	0	0
40	6	3	0	1	0
2	1	0	0	0	-1

INSERISCO x_1^a

FASE 1:

0	0	0	0	0	0	1
60	6	7	1	0	0	0
40	6	3	0	1	0	0
2	1	0	0	0	-1	1

	-2	-1	0	0	0	1	0
x_3	60	6	7	1	0	0	0
x_4	40	6	3	0	1	0	0
x_1^a	2	1	0	0	0	-1	1

$B = \{A_3, A_4, A_6\}$

$x = (0, 0, 60, 40, 0, 2)$

$\alpha = (0, 0)$

$$\theta_{MAX} = \min\left\{10, \frac{40}{6}, 2\right\} = 2$$

	0	0	0	0	0	0	1
x_3	48	0	7	1	0	6	-6
x_4	28	0	3	0	1	6	-6
x_1	2	1	0	0	0	-1	1

$B = \{A_3, A_4, A_1\}$

$x = (2, 0, 48, 28, 0, 0)$

$\beta = (2, 0)$

FASE 2:

0	-1	-1	0	0	0	1

	48	0	7	1	0	6	-6
	28	0	3	0	1	6	-6
	2	1	0	0	0	-1	1

	2	0	-1	0	0	-1	2
x_3	48	0	7	1	0	6	-6
x_4	28	0	3	0	1	6	-6
x_1	2	1	0	0	0	-1	1

$$\theta_{max} = \min\left\{\frac{48}{7}, \frac{28}{3}\right\} = \frac{48}{7}$$

	$\frac{62}{7}$	0	0	$\frac{1}{7}$	0	$-\frac{1}{7}$	$\frac{8}{7}$
x_2	$\frac{48}{7}$	0	1	$\frac{1}{7}$	0	$\frac{6}{7}$	$-\frac{6}{7}$
x_4	$\frac{52}{7}$	0	0	$-\frac{3}{7}$	1	$\frac{24}{7}$	$-\frac{24}{7}$
x_1	2	1	0	0	0	-1	1

$B = \{A_2, A_4, A_1\}$

$x = \left(2, \frac{48}{7}, 0, \frac{52}{7}, 0, 0\right)$

$r = \left(2, \frac{48}{7}\right)$

$$\theta_{max} = \min\left\{\frac{48}{7}\cdot\frac{7}{6}, \frac{52}{7}\cdot\frac{7}{24}\right\} = \left\{\frac{48}{7}, \frac{52}{24}\right\}$$

	$\frac{55}{6}$	0	0	$\frac{1}{8}$	0	0	1
x	5	0	1	[illegible]	0	0	0

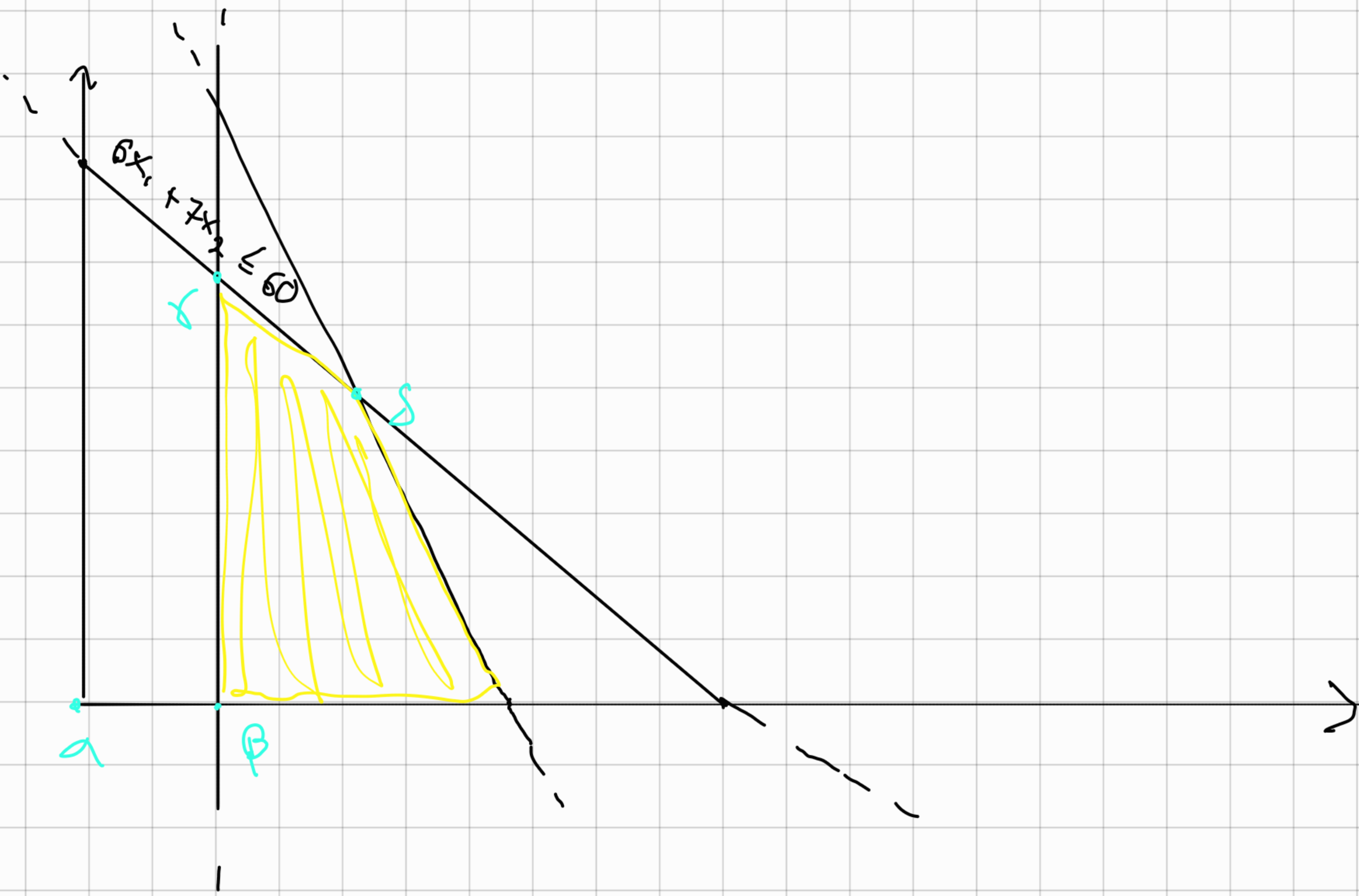
$B = \{A_2, A_5, A_1\}$

$x = \left(\frac{175}{42}, 5, 0, 0, \frac{91}{112}, 0\right)$

x_2	5	0	1	$\frac{\ }{4}$	0	0	0
x_5	$\frac{13}{6}$	0	0	$-\frac{1}{8}$	7	1	-1
x_1	$\frac{25}{6}$	1	0	$-\frac{1}{8}$	0	0	0

$\delta = \left(\frac{25}{6}, 5\right)$

$6x_1 + 7x_2 \leq 60$
$6x_1 + 3x_2 \leq 40$
$x_1 \geq 2$
$x_1, x_2 \geq 0$



KNAPSACK 0-1

$n = 4$

$p' = (13, 12, 9, 6) \quad w' = (9, 6, 8, 5) \quad C = 17$

ORDINO PER $\frac{p}{w}$:

$p' = (12, 13, 6, 9) \quad w' = (6, 9, 5, 8)$

$M_0 = \{(0,0)\}$

$M_1 = \{(0,0), (12,6)\}$

$M_2 = \{(0,0), (12,6), (13,9), (25,15)\}$

$M_3 = \{(0,0), (12,6), (13,9), (25,15), (6,5), (18,11), (19,14), \}$

$M_4 = \{(0,0), (12,6), (13,9), (25,15), (6,5), (18,11), \cancel{(19,14)}, (9,8), (21,14), \cancel{(15,13)}\}$

MIGLIOR STATO: $(25, 15) \quad x = (1, 1, 0, 0)$